

Scaling, Fractal Dynamics and Critical Exponents: Application in a non-integer dimensional ising model

Henrique A. de Lima,^{1,*} Ismael S. S. Carrasco,^{1,†} Marcio Santos,^{1,2,‡} and Fernando A. Oliveira^{1,§}

¹*International Center of Physics, Institute of Physics,
University of Brasilia, 70910-900, Brasilia, Federal District, Brazil*

²*Instituto de Física, Universidade Federal de Santa Catarina, Florianópolis, SC, Brazil*

In this work, we investigate the fundamental physics and mathematics underlying phase transitions, emphasizing the deep interplay among scaling behavior, critical exponents, and fractal geometry. In particular, we revisit the correlation function and demonstrate that applying modern fractional differentials to a fractal space of dimension d_R enables us to recover its proper form. We show that the Fisher exponent η satisfies the relation $\eta = d - d_R$, where d is the lattice space dimension and d_R is the fractional Riesz dimension, which is distinct from the fractal dimensions typically derived from percolation theory. Additionally, we examine some scaling relations that have been questioned in certain magnetic systems. Finally, considering previous results from the Ising model, we confirm that this relation holds even when d is non-integer.

Introduction - Since Mandelbrot's pioneering work [1], it has become evident that fractals play a fundamental role in physics, particularly in condensed matter and statistical physics. Near phase transitions, concepts such as renormalization and scaling are connected [2]. The scaling behavior observed at critical points is a manifestation of fractal geometries [3–5], highlighting the deep connections between scaling, renormalization, and fractals. In this work, we further strengthen these connections, clarifying concepts such as the applicability and limitations of mean-field theory.

A key quantity that deserves in-depth analysis is the correlation function

$$G(r) = \langle \psi(\mathbf{r} + \mathbf{x})\psi(\mathbf{x}) \rangle, \quad (1)$$

where $\psi(\mathbf{x})$ represents the fluctuation of the order parameter at the position $\mathbf{x}$, and $\langle \dots \rangle$ denotes the average over the lattice in a real space with dimension d . Within the mean field framework, $G(r)$ satisfies the equation

$$(-\nabla^2 + \rho^{-2})G(r) = \delta^{(d)}(r) \quad (2)$$

where ρ is the correlation length [2]. This correlation length diverges as the temperature T approaches the critical temperature T_c , following a well-known scaling behavior

$$\rho \propto |T - T_c|^{-\nu}. \quad (3)$$

The correlation function is given by $G(r) = r^{2-d} \exp(-r/\rho)$. However, this mean-field solution breaks down at the critical point. To account for this discrepancy, an empirical correction is applied [6], introducing the Fisher exponent η ,

$$G(r) \propto \begin{cases} r^{2-d} \exp(-r/\rho), & \text{if } r > \rho, \\ r^{2-d-\eta}, & \text{if } r \ll \rho. \end{cases} \quad (4)$$

Trying to describe the origins of this correction, in a recent work, Lima et al [7] conducted a thorough analysis of the correlation function (2). The authors drew inspiration from three key observations:

1. **Fluctuation-Dissipation Theorem (FDT):** The correlation functions represent a simple form of the FDT (see the article collection [8] and the review [9]), which are directly related to susceptibility [10]. The FDT is known to break down when ergodicity is violated [11, 12], a phenomenon widely studied in structural glasses [13–18], random-exchange Heisenberg chains [19], proteins [20], KPZ dynamics [21, 22], mesoscopic radioactive heat transfer [23, 24], and ballistic diffusion [11, 25];
2. **Response Functions in Growth Phenomena:** Significant progress has been made in understanding response functions in growth phenomena by assuming an underlying fractal dynamics [26–32];
3. **Scaling and Fractal Geometries:** The relation between the order parameter exponent β and the fractal dimension of the ordered phase d_f was first proposed by Suzuki [3]

$$d_f = d - \frac{\beta}{\nu}. \quad (5)$$

For the critical point of the percolation model, this fractal structure corresponds to the infinite percolating cluster [33, 34], while for general systems it is associated with the largest ordered cluster [4].

Building on these concepts, the authors analyzed the correlation function within the following basic ideas

1. Mean field theory is not efficient for analyzing the order parameter, because the fluctuations are suppressed and are strongest at the transition. However, by definition, the correlation function Eq. (2) has fluctuations at all orders. Consequently, this function should give a precise description if the right geometry is used.

* henrique.adl@hotmail.com

† ismael.carrasco@unb.br

‡ marcio.santos@ufsc.br

§ faoliveira@gmail.com

2. As $T \rightarrow T_c$ and ρ diverge according Eq. (3), the correlation extends over a large sample, exhibiting a fractal structure, which requires the use of a fractal Laplacian.
3. Thus, we replace Eq. (2) by

$$(-\nabla^2)^\zeta G(r) = \delta^{(d_R)}(r), \quad (6)$$

where d_R is a fractal dimension associated with the Riesz fractional derivative of order ζ .

4. In the context of fractal differential geometry, the Fisher exponent is derived exactly as [7]

$$\eta = d - d_R. \quad (7)$$

Consequently, the Fisher exponent in the correlation function $G(r)$, represents the deviation of the correlation fractal dimension d_R which is the same Riesz fractional dimension, from the Euclidean integer dimension d . As demanded by the Riesz fractional derivative [35, 36] $d - 1 \leq d_R \leq d$. This is consistent with the fact that $\eta \geq 0$. Furthermore, they established the relationship:

$$d_R = 2(d_f - 1). \quad (8)$$

Given the importance of such exact relationships, our goal for this manuscript is:

1. Generalize these results to non-integer dimensions d ;
2. Verify these results in extreme cases where scaling relationships break down.

Non-integer dimensional lattice - Retracing the derivation of Eqs. (7) and (8), we conclude that these relationships remain valid for non-integer dimensions d . However, fractional dynamics is subtle, particularly those related to fractional differential geometry. Thus, to check if those relationships really work for non-integer dimensions, we will analyze existing data in the literature. We will start with the Ising model in a lattice with non-integer dimension in the range $1 \leq d \leq 4$.

Before proceeding with our numerical analysis, we draw attention to the fractional exponents obtained in percolation clusters, as described, for instance, by Suzuki [3] and Coniglio[37]. The exponent d_R introduced here represents a new fractal dimension, distinct from those previously reported [3, 4, 37]. However, Eq. (8) established a connection between them.

In Table I, we present the values of d_f and d_R for the d dimensional Ising model, along with the corresponding critical exponents. The exponents β , ν and η^* are obtained from the literature, while d_f is calculated from Eq. (5), d_R from (8) and η from (7). For $d = 4$, we consider mean-field exponents, and for $d = 3$, we use the results of Pelissetto and Vicari [38].

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The computed values of d_f agree with the exact results of Coniglio [37] for $d = 2$, and 4. For $d = 1$, we use the values from the near-planar interface model [39] and the droplet model [40, 41]. For non-integer dimensions in the range $1 < d < 4$, we use β and ν from [42] and calculate d_R through Eq. (5). The values of η obtained based on this method begin to deviate from the reference [42] as d decreases.

The previous results support the extension of Eqs. (7) and (8) to non-integer dimensions. Furthermore, a detailed analysis of the data in the range $1 < d < 2$ reveals that precision decreases together with dimension, particularly for the value of η^* . However, the prediction from Eq. (7) leads to a much higher precision, highlighting the utility of this equation.

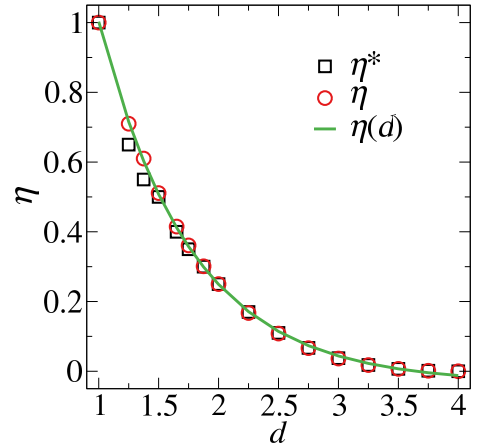


FIG. 1. Comparison between the values of η^* from the literature and the values η calculated using Eq. 7 for dimensions in the range $1 < d < 4$. All data points are taken from Table I, while the green curve represents the proposed function $\eta(d)$ given by Eq. (9).

From Table I we adjust the function

$$\eta(d) = \frac{4-d}{5d-2} \left[1 - \frac{1772(d-1)(d-2)}{625(5d-2)} \right]. \quad (9)$$

where we use the values of η for integer dimensions, being it approximate for $d = 3$ and exact for $d = 1, 2, 4$. These values are common to both η and η^* .

In Fig. 1, we present the data from Table 1 to facilitate a comparison between the values of η^* from the literature and the η calculated using Eq. (7). The graph highlights the overall agreement between these values. Additionally, the empirical function given by Eq. (9) is shown, which agrees well with the data points for $d \geq 2$, particularly with the values predicted by Eq. (7). For $1 < d < 2$ the computed values η and the function $\eta(d)$, Eq. (9), agree with each other, while they deviate slightly from

d	β	ν	η^*	d_f	d_R	η
4	1/2	1/2	0	3	4	0
3.75	0.458355(2)	0.523405(15)	0.001435(1)	2.87428(2)	3.74856(4)	0.001435(4)
3.5	0.41600(2)	0.55215(2)	0.00685(2)	2.74658(1)	3.49316(2)	0.00683(2)
3.25	0.3723(1)	0.5873(1)	0.0180(1)	2.61608(6)	3.2321(1)	0.0178(1)
3	0.3270(15)	0.6310(15)	0.0375(25)	2.481(1)	2.963(3)	0.036(3)
2.75	0.2800(30)	0.686(3)	0.067(6)	2.341(3)	2.683(5)	0.066(5)
2.5	0.2305(3)	0.758(5)	0.11(1)	2.195(2)	2.391(3)	0.108(3)
2.25	0.1790(25)	0.857(6)	0.17(1)	2.041(2)	2.082(4)	0.167(4)
2	1/8	1	1/4	15/8	7/4	1/4
1.875	0.097(3)	1.10(1)	0.30(3)	1.786(2)	1.573(4)	0.301(4)
1.75	0.068(6)	1.23(3)	0.35(5)	1.695(3)	1.389(7)	0.361(7)
1.65	0.045(10)	1.37(7)	0.40(10)	1.617(5)	1.23(1)	0.41(1)
1.5	0.01(15)	1.67(20)	0.50(15)	1.494(8)	0.99(2)	0.51(2)
1.375	-0.02(3)	2.1(5)	0.55(25)	1.38(1)	0.79(3)	0.61(3)
1.25	-0.05(5)	3.0(15)	0.65(35)	1.26(2)	0.53(5)	0.71(5)
1	0	∞	1	1	0	1

TABLE I. Values of critical exponents for the d dimensional Ising model universality class. β , ν and η^* from refs [37–42], d_f from Eq. (5), d_R from Eq. (8) while η from Eq. (7).

d	α	γ	Σ^*	Σ
2	0	1.75	2	2
1.875	-0.0625(1)	1.862(1)	1.993(7)	2.00(3)
1.75	-0.15(5)	1.99(4)	1.97(5)	2.00(2)
1.65	-0.3(1)	2.11(8)	1.93(6)	2.00(3)
1.5	-0.5(3)	2.35(2)	1.89(4)	2.0(3)
1.375	-0.9(7)	2.6(4)	1.67(25)	2.0(1)
1.25	-1.75(1.85)	3.0(1)	1.15(1)	2.0(3)
1	$-\infty$	∞	?	2

TABLE II. Values of critical exponents for the d dimensional Ising model universality class. β , γ and ν from [42] to obtain α , from Eq. (10), $\Sigma^* = \alpha + 2\beta + \gamma$ and $\Sigma = \alpha + 2\beta + \gamma$ from our values.

η^* . This suggests that the proposed function serves as a reliable empirical rule for predicting the value of the exponent η for dimensions in the range $1 \leq d \leq 4$.

Scaling - Scaling is one of the simplest and most general mechanisms in physics. In the context of phase transitions, it enables us to obtain relationships between critical exponents, providing $N - 2$ scaling relations for each set of N exponents. For instance, the critical exponent of the correlation length ν is related to the critical exponent of specific heat α through the hyperscale relation [2]

$$\alpha = 2 - d\nu, \quad (10)$$

thus associating a thermodynamic variable with the divergence of the correlation length. We have as well the Fisher scaling relation [6]

$$\gamma = (2 - \eta)\nu, \quad (11)$$

and the Rushbrooke equality [2]

$$\alpha + 2\beta + \gamma = 2. \quad (12)$$

In the previous work [7], to obtain the relation (8), some of us have used Eq. (5) along with the scaling relations above. While the Rushbrooke equality has been widely validated in equilibrium phase transitions, it was recently questioned by Fytas *et al* for the random-field Ising model [43]. If this equality fails, Eq. (8) would also break down, however, Eq. (7) would remain valid. Given that deviations may occur in the region $1 < d < 2$, we analyze these values in Table II.

In Table II, we use the values of β , γ and ν from [42] to calculate α using Eq. (10) and $\Sigma^* = \alpha + 2\beta + \gamma$. Alternatively, Σ is computed using the same expression, but with our results from Table I and the relation $\alpha + \gamma = 2 + (2 - d - \eta)\nu$, derived from the sum of Eqs. (10) and (11). The table clearly shows that as the dimension decreases from $d = 2$ to 1, Σ^* deviates from 2, while Σ remains constant within the error bounds. This distinction is crucial for $d = 1$, where $2 - d - \eta = 0$. Thus we take $(2 - d - \eta)\nu = 0$, before the limit $\nu \rightarrow \infty$ and we obtain

the correct result $\Sigma = 2$, resolving the indeterminacy in Σ^* .

Moreover, incorporating β from (5), yields:

$$\alpha + 2\beta + \gamma = 2 + (d + 2 - \eta - 2d_l)\nu. \quad (13)$$

Thus, relations (7) and (8) impose $d + 2 - \eta - 2d_l = 0$, thus validating (12) or vice versa. This shows that the differential fractal geometry linking the exponents is stronger, capable of producing reasonable exponents even when some scaling relation fails.

Dynamics - So far, we have discussed the phase transition at equilibrium within a perfect lattice of dimension d . Naturally, these ideas can be extended to explore various scenarios, ranging from equilibrium to a wide array of non-equilibrium processes. As we shall see in the following, the growth phenomenon is within the framework of these concepts.

In growth processes, two quantities play an important role in growth, the average height $\langle h(t) \rangle$ and the standard deviation of the height, also referred to as roughness or surface width, defined as $w(L, t) = [\langle h^2(t) \rangle - \langle h(t) \rangle^2]^{1/2}$. Here, the average is taken over space. The roughness is a very important physical quantity, as it controls many physical phenomena it [21, 26, 44–56]. For many growth processes, the roughness $w(L, t)$ increases over time until it reaches a saturated value w_s , such that $w(t \rightarrow \infty) = w_s \propto L^\chi$, where χ is the roughness exponent.

The surface roughness exponent χ can be related to the interface fractal dimension d_f through the expression [26, 57]

$$\chi = \begin{cases} 2 - d_f, & \text{if } d = 1, 2, \\ d - d_f, & \text{if } d \geq 2. \end{cases} \quad (14)$$

It is noteworthy that for $d \geq 2$, the Eq. (7), which describes equilibrium phase transitions, and Eq. (14), for out-of-equilibrium growth, are very similar. Furthermore, Eqs. (5), (7) and (14) consistently relate critical exponents to the deviation between the Euclidean dimensions d and a fractal dimension. Here, we generalize these concepts to lattices with non-integer d .

Conclusion - In this work, we employ exact differential fractional geometry to generalize the results obtained for lattices of integer dimension to lattices of non-integer dimension. We recover exactly the same law, Eq. (7), which establishes that the Fisher exponent η is the difference between d and the Riesz fractal dimension d_f , originating from the cluster structure formed at critical temperature T_c [7]. Using Table I and Fig. 1, we compute d_l , d_f and η to verify our claims. Additionally, we focus on the anomalous region $1 \leq d \leq 2$, see Table II, to test the robustness of our method and scaling relations, even in cases where ν diverges.

Contrary to the common assertion that mean-field theory describes critical behavior only for $d \geq d_c$, we argue that the correlation function when reformulated in terms of the Riesz fractal dimension d_R , can describe

phase transitions for any dimension d . Importantly, as $d - d_R$ decreases with d , the upper critical dimension corresponds to $d = d_c = d_R$, where our approach agrees with traditional Euclidean mean-field theory. Furthermore, Eqs. (4), (5), (7), and (8) remain valid even for non integer dimensions.

It is worth focusing on that Eq. (7) was obtained for any continuous order parameter $\psi(r)$ in a second-order phase transition, making its relevance to Ising models significant but not exclusive. Considering that out-of-equilibrium growth phenomena exhibit analogous rela-

tionships, we infer that a general relationship may exist, provided the fractal dimension d_f is well-defined for each case. Therefore, we propose that this procedure be rigorously tested in various systems, such as systems within a finite geometry[58–60], dynamic phase transitions [61–63] and synchronized phase transitions [64–68].

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